

6,193 from a Bernoulli random variable, a sample proportion of 0.5156 "rejects" $\theta = 0.495$ and "supports" $\theta = 0.5$ at a criterial evidence-ratio of $q = 3$.

Returning to our scenario of a normal random variable, for simplicity let us assume a prior distribution for the mean of $N(0,1)$ with a weight that counts it as 1 observation. Then the log of the evidence-ratio for an hypothetical μ is

$$L(\bar{X}, \mu, n) = (\bar{X}^2 - n(\bar{X} - \mu)^2 + \log(n))/2. \quad (15)$$

Given $\mu = 0$, n , and a target log-ratio $\log(q)$, the sample mean required to achieve this target is

$$\bar{X} = \pm \sqrt{(\log(n) - 2\log(q))/(n-1)}. \quad (16)$$

Note that this is very close to the sample mean identified in equation (5) for the target Robert BF (we are using a slightly different prior from his). For example, given $\mu = 0$, $n = 5000$, and target $q = 3$, the required sample mean is ± 0.035556 .

Symmetry permits us to deal with just the positive mean, so we do so from here on. Given n , target q , and the positive version of the sample mean defined in equation (16), we may solve equation (15) to find the hypothetical means required for the target to be $1/q$, which yields

$$\mu = \sqrt{\frac{\log(n) - 2\log(q)}{n-1}} \pm \sqrt{\frac{2\log(q)(n-2) + n\log(n)}{n(n-1)}}. \quad (17)$$

The left-hand term is the sample mean from equation (16), so the pair of μ in this equation always straddles the sample mean in equation (16). It also is clear that the right-hand term is larger than the left-hand term in equation (17), so the pair of μ also always straddles 0, our null-hypothesis value. Their difference is twice the right-hand term, and as n goes to infinity the right-hand term goes to 0. Thus, the pair of μ may be made arbitrarily close to one another and therefore arbitrarily close to the null that they straddle. That is, we may conclude

that for any criterial evidence-ratio level q , it is possible for a hypothetical μ_0 to be supported at that level with another μ arbitrarily close to it that is rejected at the same level.

Is a better approach available? The nub of the difficulties revealed in this paper is the use of a point-valued null. If we must make a decision for or against the null, we could a priori select a small range around the null, as in Kruschke's recommendation¹² (Kruschke, 2011, 2018) a "region of practical equivalence" (ROPE). However, here we put it to a somewhat different use from his. We may use evidence-ratios $ER(\theta, x)$ to determine support and rejection regions and ascertain whether these lie within the ROPE. Considerations regarding the width and location of the ROPE are reviewed in Kruschke (2018).

If a support region containing the null hypothesis θ_0 also is entirely contained in a ROPE around the null hypothesis value θ_0 , then we may conclude in favor of the null. In our Gaussian example, $ER(\mu, x) \geq 3$ when $0 \leq \mu \leq 0.07111$. If prior to obtaining our data we had decided that $-0.1 \leq \mu \leq 0.1$ constitutes our ROPE and our threshold is $q = 3$, then the resulting support region includes 0 and is contained by the interval $[-0.1, 0.1]$, and we therefore could decide in favor of the null. If we want to be more cautious, we could also insist on additional conditions. For instance, we could require that the average $ER(\mu, x)$ within the ROPE exceed some threshold. Conversely, if one of the rejection regions entirely contains the ROPE then we can reject the null.

Conclusions

It seems wise to take the problems raised here seriously. They demonstrate that, depending on the nature of the alternative being compared to a point-valued null, it is possible for the same data to simultaneously constitute evidence for and against the null. To review the main findings in this paper:

1. For any BF threshold q and sample mean $\bar{X}$, there is a value n such that sample sizes larger than n guarantee the occurrence of Lindley cases: Although the BF comparing H_0 against a conventional (vague) alternative exceeds q , nevertheless for some range of hypothetical μ , a BF comparing H_0 against μ in that range falls below $1/q$.
2. For any such $\bar{X}$ and q , as n increases the $BF < 1/q$ range for μ will expand and the conventional BF also will increase. The evidence for and against the null will become stronger.
3. Although the threshold $\bar{X}$ moves closer to 0 with increasing n , the upper bound on the conventional BF for which Lindley cases occur nevertheless increases.
4. For any value of q regarded by many researchers as "substantial", the probability of Lindley cases is non-negligible even when H_0 is true, and increases with sample size.
5. In a support interval setting, for any level of evidence-ratio, with increasing n it is possible for a point-null $\mu = 0$ to be supported at that level with another μ arbitrarily close to it that is rejected at the same level.
6. Bayes factors comparing a point null against point alternatives are proportional to evidence-ratios for the point alternatives. Therefore, any criterion for determining a support or rejection region will correspond to a decisional criterion for a Bayes factor comparing a point null with a point alternative, and vice-versa.

The sample sizes used here for demonstrations are not outlandish, especially in this age of Big Data. Researchers working with large data-sets are likely to encounter Lindley cases if they find small effects. The results presented here suggest that, on its own, a Bayes factor comparing a point-valued null to a vague alternative may be of little utility in assessing the degree of support for a pointwise null hypothesis, and its utility declines with increasing sample size. They also indicate that using support and rejection regions for deciding whether to reject or confirm a point null can be problematic. Here, the nature of the prior will have an